LAB 3

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3.3 Exercises

1. Code the variables definition and the process for each of the following:

a) A floating point named price initialized to 9.90.

= double price = 9.90;

b) An integer named quantity.

= int quantity;

c) A constant integer named DISC initialized to 10.

= const int DISC = 10;

d) A floating point name totalprice. Get the quantity purchased from the user and then, calculate the totalprice using the following formula: totalprice = (price*quantity) * (100-DISC)/100.00;

```
// Question 1
#include <iostream>
#include <iomanip> // i want to use setprecision
using namespace std;

int main()
{
    double price = 9.90;
    int quantity;
    const int DISC = 10;
    double totalprice;

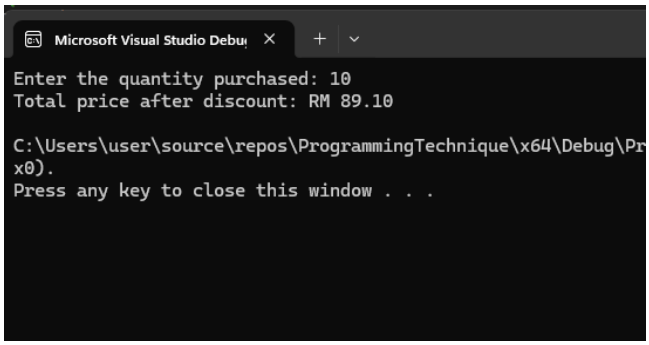
    cout << "Enter the quantity purchased: ";
    cin >> quantity;

    totalprice = (price * quantity) * (100 - DISC) / 100;

    cout << fixed << setprecision(4);
    cout << "Total price after discount: RM " << totalprice << '\n';

    return 0;
}
```

Output :



2. Write a program that prompts the user to enter the total time in seconds and displays the hours, minutes and seconds for the total time. Declare two constant integers named SCND_TO_HOUR initialized to 3600 and SCND_TO_MINUTE initialized to 60. For example, 5000 seconds are equal to 1 hour, 23 minutes and 20 seconds. Here is a sample run:

> Input the total time in seconds : ***5000***
> 5000 seconds are equal to 1 hours, 23 minutes and 20 seconds.

```
// QUESTION 2 //

#include <iostream>
using namespace std;

int main()
{
    const int SCND_TO_HOUR = 3600;
    const int SCND_TO_MINUTE = 60;
    int totalSeconds, hours, minutes, seconds;

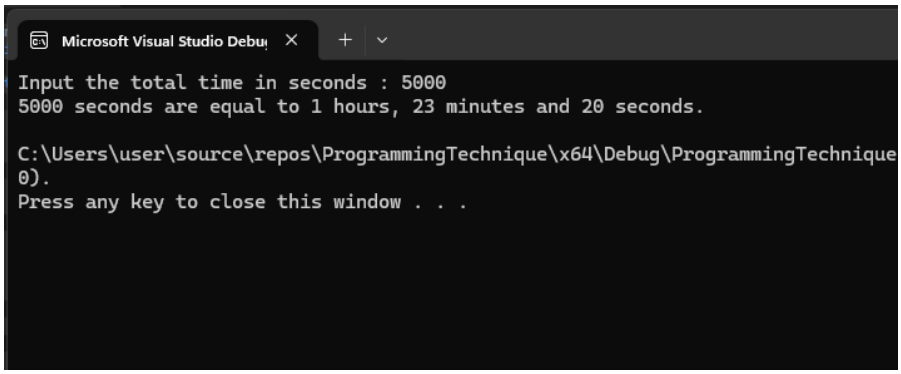
    cout << "Input the total time in seconds : ";
    cin >> totalSeconds;

    // Calculate hours, minutes, and seconds
    hours = totalSeconds / SCND_TO_HOUR;
    minutes = (totalSeconds % SCND_TO_HOUR) / SCND_TO_MINUTE;
    seconds = totalSeconds % SCND_TO_MINUTE;

    cout << totalSeconds << " seconds are equal to " << hours << " hours, " <<
minutes << " minutes and " << seconds << " seconds.\n";

    return 0;
}
```

Output :



3. Write a program that reads in loan amount, annual interest rate and number of years for the loan and display monthly payment the customer should pay for the loan. Here is a sample run :

Input loan amount (RM): ***30000***
Input annual interest rate (%) : ***4.2***
Input number of years : ***10***
Monthly payment for this loan is (RM): ***355***

```cpp
// QUESTION 3 //

#include <iostream>
using namespace std;

int main() {
    double loanAmount, annualInterestRate;
    int years;

    cout << "Input loan amount (RM): ";
    cin >> loanAmount;

    cout << "Input annual interest rate (%) : ";
    cin >> annualInterestRate;

    cout << "Input number of years : ";
    cin >> years;

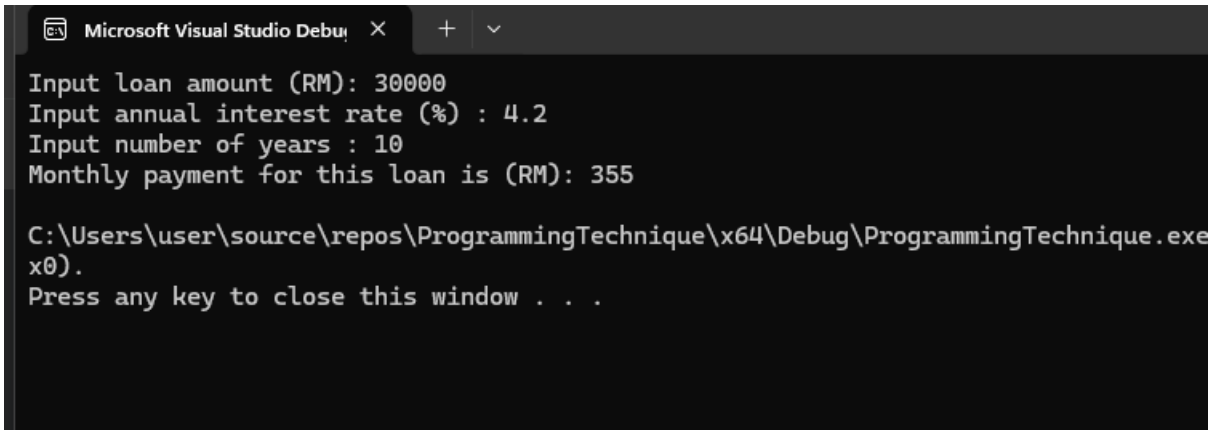
    // Calculate total interest and total payment
    double totalInterest = loanAmount * (annualInterestRate / 100.0) * years;
    double totalPayment = loanAmount + totalInterest;
    int totalMonths = years * 12;

    // Monthly payment (cast/round to integer or display directly as requested)
    int monthlyPayment = totalPayment / totalMonths;

    // Output result
    cout << "Monthly payment for this loan is (RM): " << monthlyPayment << endl;

    return 0;
}
```

Output :



4. In physics, an object that is in motion is said to have kinetic energy. The following formula can be used to determine a moving object's kinetic energy: KE = (1/2 mv2) Where KE =the kinetic energy m = the object's mass in kilograms v = the object's velocity in meters per second. Write a program to calculate the amount of kinetic energy that the object has when given the object's mass (in kilograms) and velocity (in meters per second). Here is a sample run:

Input the object's mass (kg):***10.0***
Input the object's velocity (m/s): ***5***
The kinetic energy (Joules): ***125***

```
// QUESTION 4 //

#include <iostream>
using namespace std;

int main() {
    double mass, velocity, kineticEnergy;

    // Input values
    cout << "Input the object's mass (kg):";
    cin >> mass;

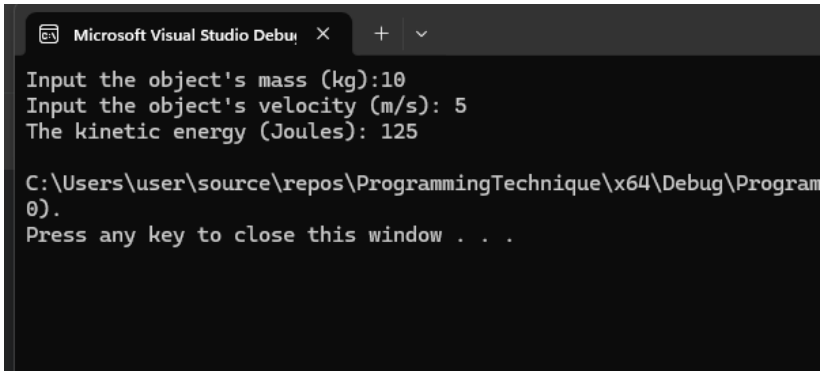
    cout << "Input the object's velocity (m/s): ";
    cin >> velocity;

    // Formula: KE = 0.5 * m * v^2
    kineticEnergy = 0.5 * mass * velocity * velocity;

    // Output result
    cout << "The kinetic energy (Joules): " << kineticEnergy << endl;

    return 0;
}
```

Output :


Microsoft Visual Studio Debu
Input the object's mass (kg):10
Input the object's velocity (m/s): 5
The kinetic energy (Joules): 125
C:\Users\user\source\repos\ProgrammingTechnique\x64\Debug\Program
0).
Press any key to close this window . . .